

Language breakthroughs

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◀ 114–115 Software errors

◀ 118–119 What do programming languages do?

For many tasks, high-level languages are better than machine code or assembly language. Two early programming languages, Fortran and BASIC, helped convince people of this.

Fortran

Short for “Formula Translation”, Fortran was developed in 1957 by a team at IBM, led by American computer scientist John Backus (1924–2007). Unlike earlier compilers, Fortran’s compiler produced machine code that ran almost as fast as hand-written code. Early Fortran programs were transformed a line at a time into patterns of holes on punched cards.

▷ Selling points

Fortran’s main selling point was that it made writing programs much easier because its syntax was much closer to English when compared with assembly languages.

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C AREA OF A TRIANGLE - HERON'S FORMULA
C INPUT - CARD READER UNIT 5, INTEGER INPUT
C OUTPUT -
    READ(5,501) A,B,C
501 FORMAT(3I5)
    IF(A.EQ.0 .OR. B.EQ.0 .OR. C.EQ.0) STOP 1
    S = (A + B + C) / 2.0
    AREA = SQRT( S * (S - A) * (S - B) * (S - C) )
    WRITE(6,601) A,B,C,AREA
601 FORMAT(4H A= ,I5,5H B= ,I5,5H C= ,I5,8H AREA= ,F10.2,
    $13H SQUARE UNITS)
    STOP
    END
```

What is it for?

Fortran is mainly used for writing programs involving scientific and mathematical problems. It was the first language to have in-built support for mathematical concepts such as complex numbers, used in many areas of physics. Fortran has been used for systems investigating nuclear physics, quantum mechanics, and the operation of aeroplanes and wind turbines.



◁ Scientific systems

Fortran is still in use today. Many scientific systems use code that was written decades ago but has proved to be very reliable over time.

Fortran is also used for weather prediction systems.

BIOGRAPHY

Grace Murray Hopper



American mathematician and Rear Admiral in the US Navy, Grace Hopper (1906–1992) was involved in developing COBOL, a programming language for businesses. She developed one of the first compilers, and her idea of making programming languages more like English helped spread computer usage.